

# Project Title

Analysis and Prediction of Product Success Using Marketing and Performance Data

## Dataset Details

For this project, I am using the Marketing and Product Performance Dataset available on Kaggle.

Source: Kaggle

Link: <https://www.kaggle.com/datasets/utkarshshrivastav07/product-sales-and-marketing-analytics-dataset>

Dataset Size: 500 rows and 15 columns

This dataset contains information about products such as price, discount, marketing spend, ratings, sales, and other performance-related features.

## Target Variable

The target variable used in this project is:

Success\_Percentage

This represents how successful a product is, and it will be used as the output variable for prediction.

## Type of Task

This project is mainly a regression task, since the goal is to predict a continuous value (success percentage).

## Problem Statement

In today's market, companies spend a lot on marketing, pricing strategies, and improving product quality. However, it is not always clear how these factors actually affect product success.

The main objective of this project is to understand how different factors such as marketing spend, price, discount, ratings, and supply chain efficiency influence product success. In addition, a machine learning model will be built to predict the success percentage of a product based on these features.

# Research Questions

- I. How does marketing spend affect product success percentage?
- II. What is the relationship between price, discount, and monthly sales?
- III. Which product categories perform the best in terms of success and sales?
- IV. How do customer ratings and number of reviews influence product success?
- V. Does supply chain efficiency have an impact on sales performance?
- VI. How do market trends and seasonality affect product success?
- VII. Which machine learning model gives the best prediction for product success percentage?

## Proposed Methodology

First, the dataset will be loaded and explored to understand its structure and features. Any unnecessary columns such as index columns will be removed. Missing values, if present, will be handled properly.

Next, basic preprocessing will be done. This includes converting data types and creating new features where required. For example, some numerical columns like marketing spend, price, and ratings may be divided into categories such as low, medium, and high.

After preprocessing, exploratory data analysis (EDA) will be performed. This includes generating summary statistics and visualizations to understand patterns in the data.

Then, for each research question, relevant columns will be selected and analyzed using grouping and aggregation techniques. Based on this, tables and graphs will be created.

For the final research question, machine learning models will be applied. The dataset will be split into training and testing sets. Models will be trained and evaluated to find which one performs best.

Finally, all results will be saved in the form of tables (CSV) and figures (PDF).

## Machine Learning Models to be Used

The following models will be used:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor

These models are suitable for regression problems and will help compare performance.

## Evaluation Metrics

To evaluate the models, the following metrics will be used:

RMSE (Root Mean Squared Error)

$R^2$  Score

These metrics will help measure how accurate the predictions are.

## Expected Figures and Tables

For each research question, one table and one figure will be generated:

RQ1: Scatter plot (Marketing Spend vs Success) + summary table

RQ2: Heatmap (Price vs Discount vs Sales) + grouped table

RQ3: Bar chart (Category vs Success) + category table

RQ4: Line chart (Rating vs Success) + rating table

RQ5: Box plot (Supply Chain Efficiency vs Sales) + summary table

RQ6: Line chart (Market Trends vs Success) + trend table

RQ7: Bar chart (Model comparison) + performance table

All figures will be saved as PDF files and tables as CSV files.